

Week 13-Random Effects-2

Li-Hsin Chien

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1 Random effects (Chapter 10, Faraway)

We illustrate the fitting methods using some data from an experiment to test the paper brightness depending on a shift operator described in Sheldon (1960). The pulp data frame has 20 rows and 2 columns. Data comes from an experiment to test the paper brightness depending on a shift operator.

```
data(pulp, package="faraway")
head(pulp)
```

```
##   bright operator
## 1   59.8        a
## 2   60.0        a
## 3   60.8        a
## 4   60.8        a
## 5   59.8        a
## 6   59.8        b
```

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i \in \{1, 2, 3, 4\}, \quad j = 1, \dots, 5$$

$$E(\alpha_i) = E(\varepsilon_{ij}) = 0, \quad \text{Var}(\alpha_i) = \sigma_\alpha^2, \quad \text{Var}(\varepsilon_{ij}) = \sigma^2$$

We can write:

$$Y = X\beta + Z\alpha + \epsilon$$

$$\epsilon \sim N(0, \sigma^2 I_{20 \times 20}) = N(0, R)$$

$$\alpha \sim N(0, \sigma_\alpha^2 I_{4 \times 4}) = N(0, D)$$

$$Y \sim N(X\beta, V = ZDZ^T + R)$$

1.1 Restricted Maximum Likelihood (REML) estimators

```
library(lme4)
```

```
## Loading required package: Matrix
```

```
mmod <- lmer(bright ~ 1+(1|operator), pulp)
summary(mmod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: bright ~ 1 + (1 | operator)
## Data: pulp
##
## REML criterion at convergence: 18.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.4666 -0.7595 -0.1244  0.6281  1.6012
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## operator (Intercept) 0.06808  0.2609
## Residual              0.10625  0.3260
## Number of obs: 20, groups: operator, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  60.4000    0.1494   404.2
```

$$\hat{\alpha} = E(\alpha|Y) = DZ^T V^{-1}(y - X\beta)$$

```
ranef(mmod)$operator
```

```
## (Intercept)
## a  -0.1219403
## b  -0.2591231
## c   0.1676679
## d   0.2133955
```

Compare with the fixed effects, which is the group mean minus the overall mean:

$$\hat{\alpha}_{i(\text{fixed})} = \bar{y}_{i.} - \bar{(\bar{y})..}$$

```
lmod <- aov(bright ~ operator, pulp)
(cc<-model.tables(lmod))
```

```
## Tables of effects
##
## operator
## operator
##      a      b      c      d
## -0.16 -0.34  0.22  0.28
```

The ratio

$$\frac{\hat{\alpha}_{i(\text{fixed})}}{\hat{\alpha}}$$

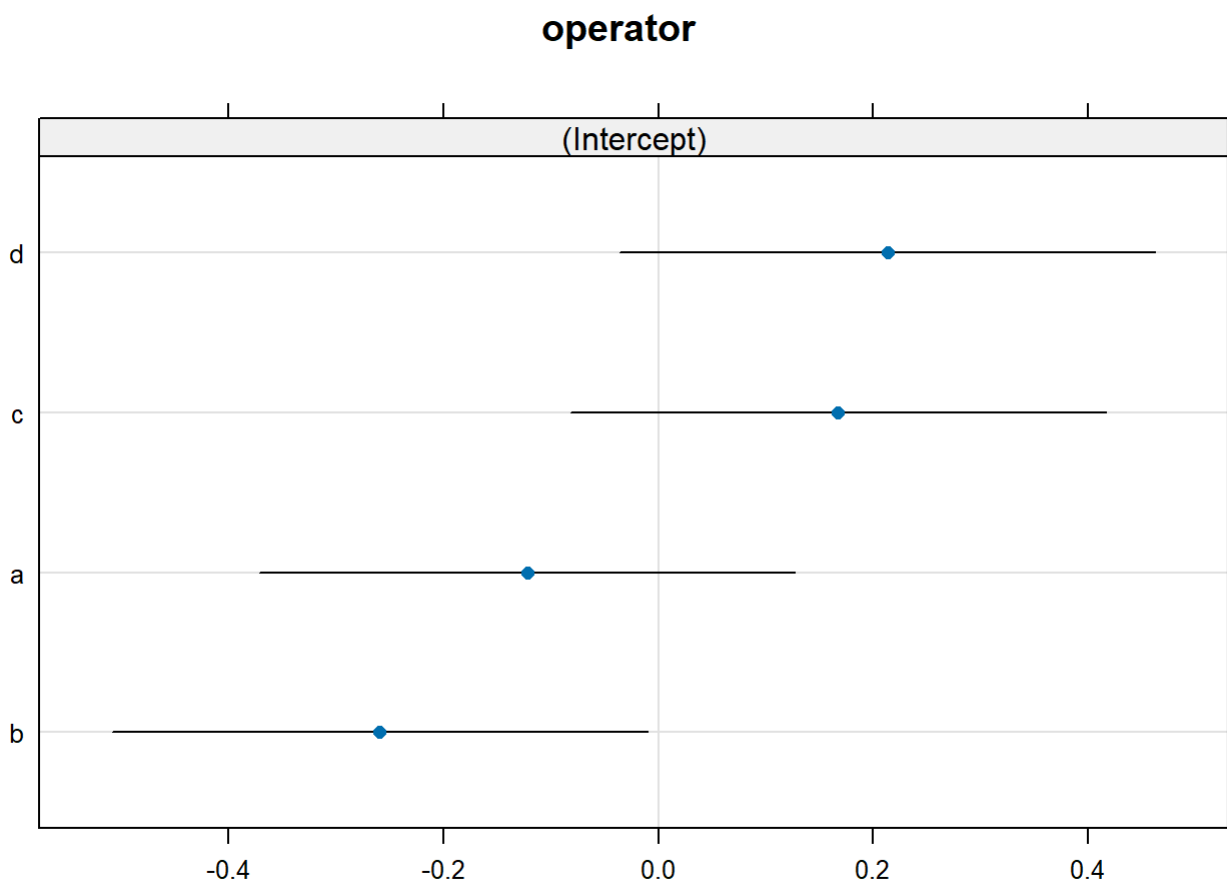
```
cc[[1]]$operator/ranef(mmod)$operator
```

```
## (Intercept)
## a 1.312117
## b 1.312117
## c 1.312117
## d 1.312117
```

The predicted random effects are exactly in proportion to the fixed effects. (shrinkage estimate)

```
library(lattice)
dotplot(ranef(mmod, condVar=T))
```

```
## $operator
```



$$Var(\alpha|Y) = D - DZ^T V^{-1} ZG$$

In the dotplot:

- points: $\hat{\alpha}_i$
- segments: approximate 95% confidence intervals for $\hat{\alpha}_i$

1.2 Prediction

Predict a new operator or unknown operator: $\hat{y}_{new} = \hat{\mu}$

Best linear unbiased predictors (BLUP) for operator $i = \hat{\mu} + \hat{\alpha}_i$

```
fixef(mmod) + ranef(mmod)$operator
```

```
## (Intercept)
## a      60.27806
## b      60.14088
## c      60.56767
## d      60.61340
```

```
predict(mmod, re.form=~0)[1]
```

```
##      1
## 60.4
```

```
predict(mmod, newdata=data.frame(operator="a"))
```

```
##      1
## 60.27806
```

Compute a confidence interval for the prediction of a new(unknown) operator using the parametric bootstrap.

$$y_{new} = \mu + \alpha_{new} + \epsilon$$

```
group.sd <- as.data.frame(VarCorr(mmod))$sdcor[1]
resid.sd <- as.data.frame(VarCorr(mmod))$sdcor[2]
pv <- numeric(1000)
for(i in 1:1000){
  y <- unlist(simulate(mmod))
  bmod <- refit(mmod, y)
  pv[i] <- predict(bmod, re.form=~0)[1] + rnorm(n=1,sd=group.sd) + rnorm(n=1,sd=resid.sd)
}
quantile(pv, c(0.025, 0.975))
```

```
##      2.5%      97.5%
## 59.47309 61.25233
```

Compute a confidence interval for the prediction of operator a using the parametric bootstrap.

$$y|\alpha_1 = \mu + \alpha_1 + \epsilon$$

```
for(i in 1:1000){
  y <- unlist(simulate(mmod, use.u=TRUE))
  bmod <- refit(mmod, y)
  pv[i] <- predict(bmod, newdata=data.frame(operator="a")) + rnorm(n=1,sd=resid.sd)
}
quantile(pv, c(0.025, 0.975))
```

```
##      2.5%      97.5%
## 59.66917 60.97733
```